Particle in a ring

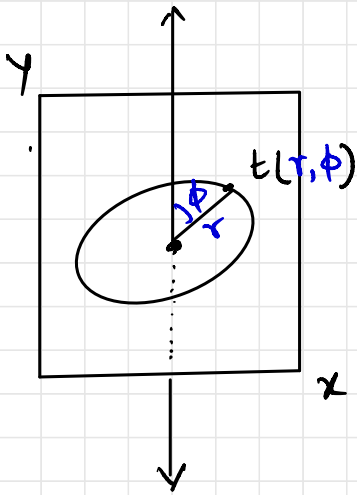
TISE

$$-\frac{\hbar^2}{2I}\frac{d^2\psi}{d\phi^2} + m_l^2\psi = 0$$

$$\left\{ E = \frac{m_l^2\hbar^2}{2I} \right\} = \frac{h^2 m_l^2}{8\pi^2 m r^2}$$

$$I = mr^2$$

Energy



Wavefunction

$$\psi(\phi) = A e^{i m_l \phi} + B e^{-i m_l \phi}$$

↓ cyclic BC

$$\psi(\phi) = \psi(\phi + 2\pi)$$

$$\psi(\phi) = \frac{1}{\sqrt{2\pi}} e^{i m_l \phi}$$

for $m_l = 0, \pm 1, \pm 2, \pm 3 \ldots$

Particle in a box

$$V(x) = 0$$

Wavefunction

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$$

$$n = 1, 2, 3, \ldots$$

Energy

$$E_n = \frac{n^2 h^2}{8 m a^2}$$

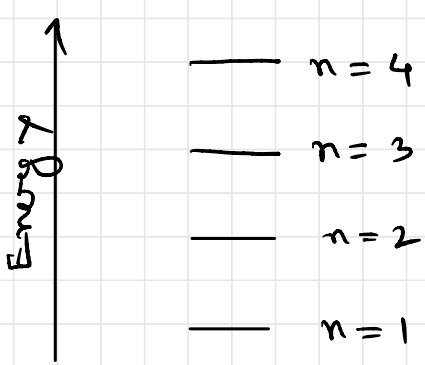
Lowest energy quantum State

$$n = 1$$

$$E_1 = \frac{h^2}{8 m a^2}$$

ZERO POINT ENERGY $\neq 0$

NO DEGENERACY



Particle in a ring

$$V(x,y) = 0$$

$$\psi(\phi) = \frac{1}{\sqrt{2\pi}} e^{i m_l \phi}$$

$$m_l = 0, \pm 1, \pm 2, \ldots$$

$$E_{m_l} = \frac{m_l^2 h^2}{8 \pi^2 m r^2}$$

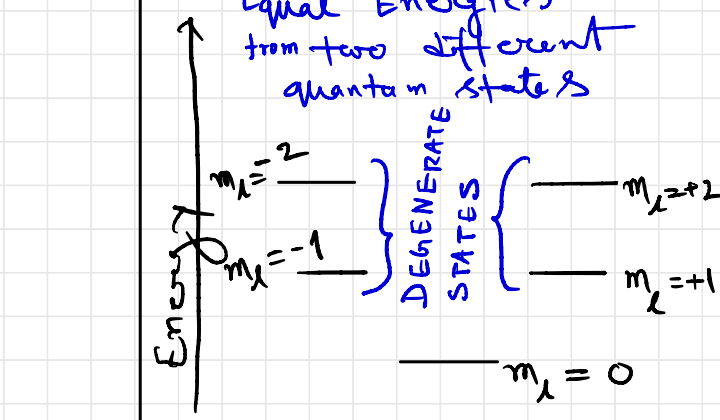
$$m_l = 0$$

$$E_0 = 0$$

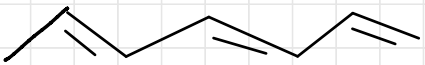
ZERO POINT ENERGY = 0

DEGENERACY

Equal Energies from two different quantum states



APPLICATION

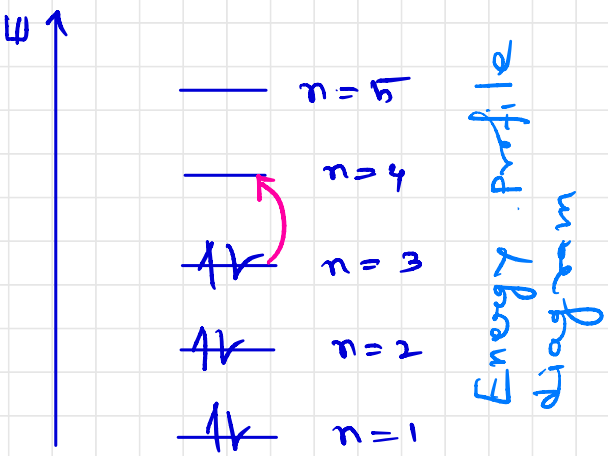


π - bonds = 3

No. of π-electrons = 3 x 2 = 6

ΔE = ?

Particle in a box-model



Occupied

$n = 3$

$E_3 = \frac{9h^2}{8ma^2}$

Unoccupied

$n = 4$

$E_4 = \frac{16h^2}{8ma^2}$

$\Delta E = E_4 - E_3$

Excitation energy from highest energy occupied quantum state to lowest energy unoccupied quantum state

3 4

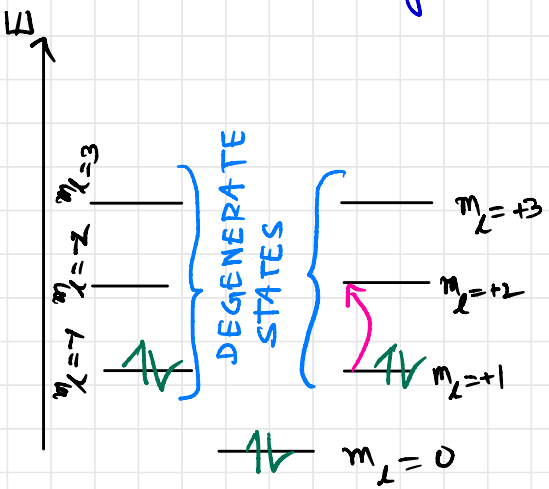


π - bonds = 3

No. of π-electrons = 3 x 2 = 6

ΔE = ?

Particle in a ring-model



Occupied

$m_l = \pm 1$

$E_1 = \frac{h^2}{8\pi^2 mr^2}$

Unoccupied

$m_l = \pm 2$

$E_2 = \frac{4h^2}{8\pi^2 mr^2}$

$\Delta E = E_2 - E_1$

Excitation energy from highest energy occupied quantum state to lowest energy unoccupied quantum state

1 2